

Offline Handwritten Bangla Character Recognition using Few-Shot Learning

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Abstract—Character recognition is becoming more and more important as a result of its numerous applications. Optical character recognition has several uses for accessibility, storability, backups, and translation among other things, in the legal, health-care, and financial sectors in the real world. This study focuses on a discussion of different character recognition methods for Bangla and other languages. A variety of handwritten character recognition methods have been developed for Bangla language. Yet, Bangla handwritten characters are difficult to recognize because of their variety, similarity, and compound characters. In a few papers, various deep learning techniques were applied to the recognition of Bangla characters. This paper's goal is to suggest a novel strategy for recognizing Bangla characters. Considering the novelty of this field, the application of a few-shot learning approach, particularly because there has been no work on Bangla character recognition using this method. This approach is also well-suited for low-resource datasets. While other languages such as Chinese, Tamil, Urdu, and Malayalam have seen some work using few-shot learning, this remains an unexplored area for Bangla character recognition.

Index Terms—Bangla handwritten character recognition, feature extraction, Few-shot learning, prototypical network, deep learning

I. INTRODUCTION

There are many people around the world who use Bangla as a language. Despite Bangla's widespread usage and popularity, there aren't as many works in it as there are in other widely used languages, including English. Because the style of the Bangla language is entirely distinct from that of English or other languages. Given that Bangla characters combine numerals, simple characters, and compound characters, Bangla's handwritten characters are distinctive and challenging. They have a different range of dimensions, forms, and sizes. Contrarily, the most well-liked and active area of pattern recognition research is optical character recognition. A technique for converting printed or handwritten text or images into digital files is optical character recognition.

This thesis research focuses on implementing and evaluating a few-shot learning approach for Bangla handwritten character recognition using a Prototypical Network integrated with various pre-trained models such as VGG16, MobileNet, ResNet50, InceptionV3, and DenseNet121. To assess their performance in recognizing Bangla characters with minimal training data. This work not only aims to enhance Bangla character recognition, but also opens the way for applying these techniques to tribal and low-resource languages. By developing effective recognition models for these languages, we can contribute to preserving linguistic diversity and ensuring that digital technologies are accessible to a broader range of tribal and unprivileged communities.

A. Research Problem

Optical character recognition (OCR) is a technique that converts the text into a format that computers can understand. In the modern world, OCR is useful for digitizing typewritten materials as well as handwritten ones. It makes it much easier to find important information in a pile of documents and files, saving a lot of manual searching.

In optical character recognition (OCR), the features of characters and categorizing them based on patterns are two important parts [1]. There's a specific area within OCR called Handwritten Optical Character Recognition, which has gained a lot of interest. This area is divided into two types: offline and online systems, depending on how the input is given. Offline systems work with static input, like scanned pictures, while online systems are more dynamic and rely on the movement of a pen, considering factors like speed, angle, position, and path. Online systems are seen as more advanced because they address the issue of input overlap, which offline systems struggle with.

With the development of technology, one of the first OCR systems was created in the 1940s [1]. With advancements, OCR technology evolved to recognize both printed and handwritten

characters. Eventually leading to OCR machines being sold commercially. A significant milestone was the introduction of the "IBM 1287" reading a device at the 1965 New York World Fair. It was the first optical reader capable of recognizing handwritten numbers.

In the past ten years [1], researchers have attempted to enhance the accuracy of OCR systems by fusing with image processing techniques. Researchers worked on various techniques like support vector machines, random forests, k-nearest neighbors, decision trees, and neural networks. In recent years, scientists have mostly concentrated on creating deep learning-based algorithms for digitizing handwritten materials.

Few-shot learning is an important and challenging area in deep learning, which focuses on learning from a limited number of labeled examples. This approach is gaining attention because deep learning typically requires large-labeled datasets, which are not always available for various reasons. Learning from a few examples is a human-like learning process. Few-shot learning has shown significant potential in various machine learning applications, especially in image classification. Since it is a relatively new technique, many researchers are exploring its foundational concepts using image datasets like Mini-ImageNet and Omniglot. Additionally, few-shot learning offers opportunities, to work with [22] low-resource languages. Prototypical networks, a popular approach to few-shot learning, are often used as a baseline in these studies. Prototypical networks are an effective method in few-shot learning, representing each class with a prototype, which is the mean of the feature embeddings from the support set. During inference, a query sample is classified by measuring its distance to these class prototypes. This approach is computationally efficient and avoids complex meta-learning strategies, making it widely used in areas like computer vision and natural language processing.

B. Research Objectives

After studying Bangla character recognition research papers and trying to figure out a novel approach in this field, we chose to work with a few-shot learning approach. Few-shot learning is a machine learning technique that can recognize new objects using a small amount of labeled data. The following is a list of the objectives of our research:

- 1) Traditional deep learning requires huge datasets, which is an obstacle for my research.
- 2) Making a research path for low-resource language datasets which have complex characteristics like the Bangla language.
- 3) Experiment with one of the few-shot approaches; here, it is the prototypical network.
- 4) To experiment how the prototypical network fits with image classification models like MobileNet, VGG16, ResNet50, DenseNet121, InceptionV3.
- 5) To find out the strengths and limitations of the few-shot approach with the prototypical network.

II. LITERATURE REVIEW

There have been many types of research conducted in this character recognition field using the machine learning approaches.

Pramanik et al. suggested a cutting-edge shape decomposition-based segmentation method for breaking down compound features in their paper [2]. This form of decomposition enhances recognition accuracy while reducing categorization difficulty by requiring fewer classes to recognize. The segmentation area, where two fundamental shapes are combined to create a compound character, is where the deconstruction takes place. The classifier employed by the authors to classify data was based on a multi-layer perceptron model and used backpropagation learning. They conducted their experiment using the CMATERdb dataset and the ICDAR 2013 Segmentation dataset. These datasets are objective and contain a variety of components in this particular field. The investigation used 10,240 photos of compound characters. For training reasons, they have employed 12,300 basic character graphics. They not only have a high recognition rate, but they also employ fewer classes to produce such precise results. Classes for compound characters no longer require separate definitions. Their research will pave the way for a novel approach to utilising the compound characters' shapes for Bangla character recognition.

The authors of this paper [3] suggested an approach that normalises the written character pictures before using CNN to classify specific characters. It does not use any feature extraction techniques. In this work, 20000 handwritten characters in various shapes and variations are employed. The author's approach performed better than several other important new approaches. For 50 isolated Bangla characters, they prepared a sizable handwritten dataset. For handwritten character scripts, they took into account about 30 people of various ages and educational levels. Their prepared dataset size is 20000 with 400 samples per character. Some of these character illustrations have extremely intricate shapes and are closely related to one another. In their paper on handwritten characters, the authors explored the efficiency of deep convolutional neural networks (DCNNs) [4]. performance on the CMATERdb dataset. The authors of this paper [5] use the SE-ResNeXt to present a deep Convolutional Neural Network model. The proposed model could identify Bangla handwritten compound characters with an average accuracy of 99.82%. Also, the suggested model exhibits higher outcomes than the most advanced models, outperforming them.

To recognise Bangla handwritten letters, Das et al. suggested an extended convolutional neural network (CNN) model [6]. Their model is evaluated using the "BanglaLekha-Isolated" dataset, which has 39 classes of consonants and 11 classes of vowels. That model indicates that for Bangla numbers, 99.50% of the identification accuracy is accurate; vowels got 93.18%, consonants got 90.00%, and combined classes got 92.25%. Three convolutional layers, one picture input layer, and three fully connected layers make up the architecture of their

suggested model. In another paper [7], the authors used face mapping to upgrade the recognition of Bangla handwritten characters. The results of their handwriting recognition were surprisingly good using a basic machine learning technique. They achieved a 94% accuracy for individual characters and an average of 91% for all characters. They focused on eleven vowel characters for better identification and used Scikit-Learn along with an additional SVM classifier on a matrix dataset for their solution.

Rizvi [8] et al. concentrated on two distinct architecture. First, a hybrid SVM classifier model trained these features. Second, they take into account ResNet, a multilayer CNN that employs residual learning and automatically picks up the information required for recognition.

The authors used two datasets, BanglaLekha-Isolated and Ekush. These datasets contain collections of distinct, handwritten Bangla characters. They utilized 700 samples from BanglaLekha-Isolated and 3000 samples from Ekush for each unique character to create their dataset. The dataset was divided into three groups: basic characters (including vowels and consonants), numbers, and compound characters. Their findings showed that ResNet-18, a CNN-based model, outperformed conventional classifier-based methods.

According to Rabby et al., main characters, digits, modifiers, and compound characters printed in Bangla handwritten characters can be recognised [9]. Because of this problem, each person writes handwritten characters in their own unique way and opts for handwritten scripts. Handwritten Bangla character can be recognised using the model Ekushnet, proposed by the authors. The Ekush dataset was used to train the suggested model, validate it, and cross-validate it using the CMATERdb dataset. Their proposed approach achieved a recognition accuracy of 97.73% on the Ekush dataset and a cross-validation accuracy of 95.01% on the CMATERdb dataset.

In this paper [10], Rabby et al. suggested a straightforward, lightweight CNN model. Using 50 fundamental Bangla characters, their model can categorise Bangla handwriting characters. The Banglalekha-Isolated, CMATERdb, and ISI datasets were used in their experiment. Their character recognition model, BomoNet, achieved validation accuracy scores of 98% on the CMATERdb dataset, 96.81% on the ISI dataset, 95.71% on the Banglalekha-Isolated dataset, and 96.40% on the mixed dataset.

In this paper, the authors conduct a comparative investigation of various character recognition methods in various languages [11]. Several languages, including English, Bangla, Marathi, Devanagari, Oriya, Chinese, Latin, and Arabic, are the focus of this analysis. Based on these variables, such as the dataset and approach employed, this paper provides a comparative comparison of machine learning, and deep learning strategies for handwritten character recognition. Due to the variety of characters in various languages, handwritten character recognition is a difficult task. Segmentation, feature extraction, and classification algorithms are the main topics of this research.

In this paper[19], Snell et al. experimented on the Omniglot and miniImageNet datasets, demonstrating that prototypical networks achieve state-of-the-art results, significantly outperforming other models. They achieved 98.8% accuracy for 1-shot 5-way classification and 99.7% accuracy for 5-shot 5-way classification on the Omniglot dataset. Similarly, on the miniImageNet dataset, they reported 49.42% accuracy for 1-shot 5-way and 68.20% accuracy for 5-shot 5-way classification. Prototypical Networks address the few-shot learning challenge by training a model to generalize to new classes with minimal examples. This is accomplished by learning a metric space where classification is determined by measuring distances to class prototypes. This approach

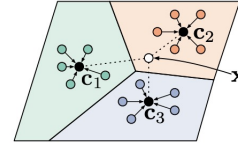


Fig. 1. Prototypical Networks in the few-shot scenario

simplifies the learning process and proves to be more effective than complex meta-learning techniques such as Matching Networks and LSTM-based models. Prototypical Networks compute class prototypes as the mean of support examples in an embedding space generated by a neural network. Episodic training, which mimics the few-shot task, enhances generalization. The model also excels in zero-shot learning, outperforming existing methods on the CUB dataset. Prototypical Networks offer a simple, efficient, and a highly effective solution for few-shot and zero-shot learning, with potentiality for further optimization by exploring different distance metrics and improving the embedding function.

In this [21] research, Sahay et al. discussed the challenges of recognizing handwritten Urdu characters using an enhanced Prototypical Network architecture for few-shot learning. Due to the limited availability of clean, annotated data for Urdu—a language that is context-sensitive and written in a bi-directional script—traditional OCR methods struggle to achieve high accuracy. The proposed solution involves a Prototypical Network that learns Euclidean embeddings of images and uses class prototypes for classification. This study demonstrates the effectiveness of Prototypical Networks in recognizing Urdu characters with minimal examples, paving the way for improved OCR models for low-resource languages.

Shaffi et al. explored the application of Deep Siamese Convolutional Neural Networks (Siamese CNNs) for Tamil handwritten character recognition within the context of few-shot[20] learning. Given the challenge of insufficient training samples in many optical character recognition (OCR) applications, the authors leveraged Siamese CNNs' capability to perform well with minimal data. Their model employs a twin CNN architecture. The Siamese CNN

achieved an optimal accuracy of 83.39% in a 40-shot learning scenario. Through extensive experimentation on the uTHCD database, which includes a diverse set of Tamil handwritten characters, the study demonstrates the model's robustness and effectiveness in recognizing characters despite the limited data availability. This research sets a new benchmark for developing efficient Indic OCR models using Siamese networks and highlights the potential for further improvements.

III. METHODOLOGY

Our purpose is to recognize Bangla characters in few shot approach with prototypical network. For that, we collected our data from Mendely data. We will experiment here how prototypical networks fit with image classification models like ResNet50, MobileNet, VGG16, InceptionV3, DenseNet121 by testing this dataset.

A. Dataset Collection

For my thesis project, I utilized the BanglaLekha-Isolated image dataset, which I downloaded from Mendeley Data. This dataset contains 84 distinct characters, including 50 basic Bangla characters, 10 Bangla numerals, and 24 selected compound characters. Each character has 2,000 handwritten samples that were collected, digitized, and pre-processed. After eliminating errors and scribbles, the dataset comprises a total of 166,105 handwritten character images. All images within each character folder are uniquely labeled.

B. Dataset Description

In few-shot learning, the main goal is to achieve better results with fewer samples. To align the dataset with this technique, I reduced its size to make it suitable for few-shot learning. I selected 50 basic Bangla characters, 10 Bangla numerals, and 12 compound characters, totaling 72 characters. For each character, I chose 20 image samples. The images for each character are organized into separate classes.

C. Data Processing

The dataset of handwritten characters is first loaded and resized to a consistent dimension of 224x224 pixels, making it suitable for input into the image classification models. Each character class is stored in separate folders, with the respective images inside. The images are then converted into numerical arrays and labeled according to their respective classes. Once loaded, the images undergo normalization, where the pixel values are scaled to the range [0, 1], which helps enhance model performance by standardizing the input data. The dataset is subsequently divided into training and testing sets, with 80% allocated for training and 20% for testing, ensuring a balanced class distribution through stratified sampling.

To implement a prototypical network for few-shot learning, the dataset is divided into a support set and a query set. Each

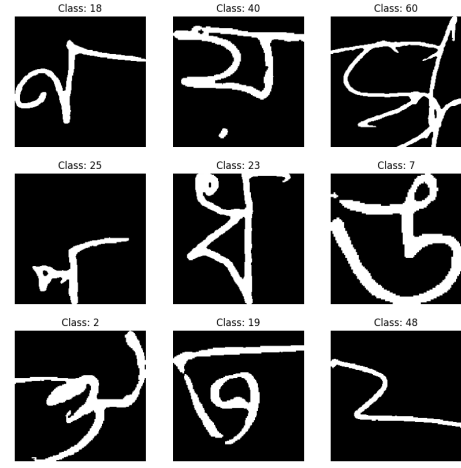


Fig. 2. Characters divided in classes

Support Set (5 Images) for Character 12



Fig. 3. Support Set

class contains 20 images, with 5 images randomly selected to form the support set, and the remaining 15 images assigned to the query set. The support set is used to calculate class prototypes, which are derived as the mean embeddings of the support images for each class. Classification is achieved by measuring the distance between the embeddings of query images and the class prototypes, with the query images classified based on the nearest prototype. This method enables the model to generalize to unseen examples using only a few samples from the support set per class.

IV. RESULT ANALYSIS

A. Models Implementation

We used various pre-trained models to train this extracted dataset. We got different accuracy as a result for different models.

B. ResNet50

ResNet50 is a deep convolutional neural network consisting of 50 layers. It employs a unique architecture called residual

Query Set (15 Images) for Character 72



Fig. 4. Query Set

learning, where shortcut connections are added to skip the layers and make it easier to train, even as the model deepens. ResNet50 is known for its ability to extract complex features from images through multiple layers of convolution, pooling, and activation functions. The ResNet50 model takes images of size 224x224 as input and output feature vectors that represent the key characteristics of the image, which can then be used for classification or other tasks. However, in the context of few-shot learning, this step is modified or replaced as part of the prototypical network setup. Here, the ResNet50 backbone is combined with a prototypical network approach. The ResNet50 backbone is used to extract embeddings for both the support set and query set images. The embeddings of the query set images are then compared to the class prototypes using a distance metric, which is Euclidean distance in this case. Each query image is assigned to the class whose prototype is closest in the feature space. Based on the distance to the prototypes, the query set images are classified.

C. MobileNet

MobileNet is a compact deep learning model optimized for image classification. It employs a simplified structure using depthwise separable convolutions, which reduces both the number of parameters and computational load, enhancing speed and memory efficiency. Despite its lightweight structure, MobileNet effectively extracts features from images, making it well-suited for transfer learning and applications like few-shot learning, where efficient feature extraction is essential. The model is built around a pre-trained MobileNet architecture, which is employed for feature extraction. The input size is set to (224, 224, 3) to match the image data, and all images are resized to this shape before processing them by the model. MobileNet is used as the base model in a new architecture, where input images pass through it to produce feature embeddings. These embeddings represent the images in a lower-dimensional space, enabling comparison in the few-shot learning task. For each class, the feature embeddings of the support images are used to compute a prototype, which is the mean vector of the embeddings. This prototype serves as the class representation in the embedding space. The class whose prototype is closest which has the smallest distance is predicted as the label for the query image. In this experimental setup, MobileNet functions as a feature extractor, generating embeddings that are input into the prototypical network for classifying Bangla characters based on their distance to class prototypes.

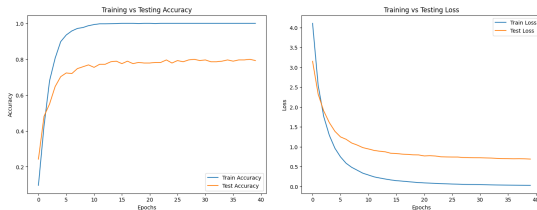


Fig. 5. MobileNet Model Training Vs Testing Accuracy and Loss

D. VGG16

The VGG16 model is a convolutional neural network commonly used for image classification tasks. It has a total of 16 layers, comprising 13 convolutional layers and 3 fully connected layers. The network employs small 3x3 filters and follows a consistent architectural pattern, which makes it direct and simple yet powerful for feature extraction. Pre-trained on the ImageNet dataset, VGG16 is often applied in transfer learning, enabling it to perform well on new tasks with limited training data. VGG16, a well-known convolutional neural network (CNN), is pre-trained on the ImageNet dataset and has been adapted to create embeddings for the Bangla handwritten character recognition problem. The VGG16 model is used as the base feature extractor. This is significant for character recognition. Its layers are frozen during training to leverage pre-learned features. A GlobalAveragePooling2D layer is added on top of the VGG16 base model, which converts the spatial feature maps into a compact feature vector. This feature vector serves as the embedding for each input image. The embeddings generated by the VGG16 model are used as inputs to the Prototypical Network to compute class prototypes and perform few-shot classification. This architecture combines the robustness of VGG16 for feature extraction with the efficiency of Prototypical Networks for handling limited training samples per class.

E. InceptionV3

The InceptionV3 model is a deep convolutional neural network designed for image classification tasks. It captures important features at multiple scales by using parallel convolutional layers with different filter sizes. The model was pre-trained on large datasets, which allows it to recognize complex patterns in images. The convolutional layers from the InceptionV3 model are utilized, and its weights are frozen to retain the pre-trained knowledge. The fully connected layers from the original architecture are removed. The Prototypical Network is employed to perform the few-shot classification. In this setup, the prototypical network leverages the embeddings generated by InceptionV3 to classify images based on their distance from prototype vectors.

F. DenseNet121

The DenseNet121 model is a deep convolutional neural network designed for image classification. It features densely connected layers, with each layer receiving input from all previous layers within the same block. This unique structure allows for efficient feature reuse, improving the flow of information and reducing the number of parameters compared to traditional architectures. The model used in this few-shot learning setup is based on DenseNet121, a deep convolutional neural network designed for image classification. In this combination, DenseNet121 extracts meaningful feature vectors from the images, and the prototypical network uses these features to perform a few-shot classification based on the calculated distances between prototypes and query embeddings.

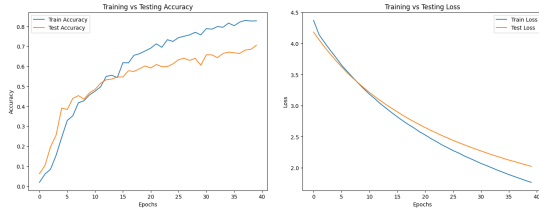


Fig. 6. DenseNet121 Model Training Vs Testing Accuracy and Loss

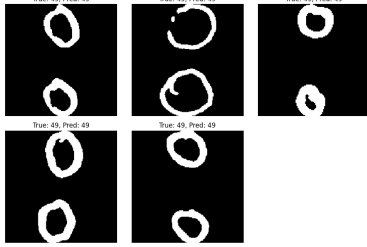


Fig. 7. Output character Prediction from Query Set

V. RESULTS IN A TABLE

The table below shows the precision, recall, F1 score, accuracy for each model respectively.

Model	Precision	Recall	F1 Score	Accuracy
ResNet50	0.70	0.68	0.66	0.68
VGG16	0.62	0.62	0.59	0.62
MobileNet	0.82	0.79	0.79	0.79
InceptionV3	0.69	0.66	0.64	0.66
DenseNet121	0.72	0.70	0.70	0.71

TABLE I
PERFORMANCE METRICS TABLE

VI. CONCLUSION

In this paper, we explored the application of few-shot learning using Prototypical Networks for the handwritten Bangla character recognition. We conducted experiments with several deep learning models, including ResNet50, VGG16, MobileNet, InceptionV3, and DenseNet121, to evaluate their performance using a minimal number of training samples. Among these models, MobileNet demonstrated the best overall performance, followed by DenseNet121 and ResNet50, with VGG16 performing the weakest. The research highlighted the potential of few-shot learning approaches, particularly in scenarios where large datasets are unavailable, which is a common issue in low-resource languages. This work creates opportunities for applying advanced recognition techniques to underrepresented languages, contributing to the development of OCR systems that are inclusive and capable of preserving linguistic diversity. It would further advance the field of character recognition in underrepresented languages, ensuring

that digital technologies become more accessible to a wider range of communities for their lifestyle convenience.

REFERENCES

- [1] J. Memon, M. Sami, R. A. Khan, and M. Uddin, "Handwritten optical character recognition (ocr): A comprehensive systematic literature review (slr)," *IEEE Access*, vol. 8, pp. 142 642–142 668, 2020.
- [2] R. Pramanik and S. Bag, "Shape decomposition-based handwritten compound character recognition for bangla ocr," *Journal of Visual Communication and Image Representation*, vol. 50, pp. 123–134, 2018.
- [3] M. M. Rahman, M. Akhand, S. Islam, P. C. Shill, M. Rahman *et al.*, "Bangla handwritten character recognition using convolutional neural network," *International Journal of Image, Graphics and Signal Processing*, vol. 7, no. 8, pp. 42–49, 2015.
- [4] M. Z. Alom, P. Sidike, M. Hasan, T. M. Taha, V. K. Asari *et al.*, "Handwritten bangla character recognition using the state-of-the-art deep convolutional neural networks," *Computational intelligence and neuroscience*, vol. 2018, 2018.
- [5] M. M. Khan, M. S. Uddin, M. Z. Parvez, and L. Nahar, "A squeeze and excitation resnext-based deep learning model for bangla handwritten compound character recognition," *Journal of King Saud University-Computer and Information Sciences*, vol. 34, no. 6, pp. 3356–3364, 2022.
- [6] T. R. Das, S. Hasan, M. R. Jani, F. Tabassum, and M. I. Islam, "Bangla handwritten character recognition using extended convolutional neural network," *Journal of Computer and Communications*, vol. 9, no. 03, pp. 158–171, 2021.
- [7] S. Ahsan, T. N. Shah, T. B. Sarwar, M. S. U. Miah, and A. Bhowmik, "A machine learning approach for bengali handwritten vowel character recognition," *IAES International Journal of Artificial Intelligence*, vol. 11, no. 3, p. 1143, 2022.
- [8] M. RIZVI, K. Deb, M. I. Khan, M. KOWSAR, M. SAKI, and T. KHANAM, "A comparative study on handwritten bangla character recognition," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 27, no. 4, pp. 3195–3207, 2019.
- [9] A. S. A. Rabby, S. Haque, S. Abujar, and S. A. Hossain, "Ekushnet: using convolutional neural network for bangla handwritten recognition," *Procedia computer science*, vol. 143, pp. 603–610, 2018.
- [10] A. S. A. Rabby, S. Haque, S. Islam, S. Abujar, and S. A. Hossain, "Bornonet: Bangla handwritten characters recognition using convolutional neural network," *Procedia computer science*, vol. 143, pp. 528–535, 2018.
- [11] A. Singh and A. S. Bist, "A wide scale survey on handwritten character recognition using machine learning," *Int J Comput Sci Eng*, vol. 7, no. 6, pp. 124–134, 2019.